Agentic Recovery for Lightweight Vision-Language-Action Policies

Author Names Omitted for Anonymous Review. Paper-ID [add your ID here]

***Abstract*—TODO.**

I. Introduction

TODO.

II. Related Work

A. Efficient Vision-Language-Action Policies

TODO cite SmolVLA [8], TinyVLA [9], OpenVLA [4], and broader robot foundation policies such as RT-1 [2] and RT-2 [11].

B. Agentic Planning and Feedback

TODO cite SayCan [1], Inner Monologue [3], Code as Policies [5], ReAct [10], and Reflexion [7]. Tie this to SemRob topics: semantic goal interpretation, policy steering, online correction, and uncertainty or failure monitoring.

C. Benchmarks and Failure-Recovery Evaluation

TODO cite LIBERO [6]. State why final success must remain the benchmark success flag rather than a wrapper-internal verifier decision.

III. Method

A. Base Policy and Evaluation Harness

TODO describe the fixed SmolVLA policy, observation/action interface, LIBERO or ManiSkill environment, seeds, action budgets, and logging artifacts.

B. Agentic Recovery Wrapper

TODO define planner, verifier, retry scheduler, and replan policy. Distinguish episode-level retry from subgoal-level in-episode recovery if both appear.

C. Success Semantics

TODO state explicitly: final task success is the environment benchmark success signal, e.g., `info["success"]` or `is_success`. Wrapper verifier predicates are used only for control decisions such as continue, retry, or replan.

IV. Experiments

A. Setup

TODO list suites, task IDs, episodes per task, compute environment, simulator versions, policy checkpoint, and action-step protocol.

TABLE I
POLICY-ONLY BASELINE REPRODUCTION ON LIBERO. FILL WITH FINAL PAPER-SCALE NUMBERS ONLY.

Condition	Goal	Object	Spatial	Long	Avg
Policy-only	TODO	TODO	TODO	TODO	TODO
Reference	TODO	TODO	TODO	TODO	TODO

TABLE II
AGENTIC RECOVERY ABLATION. KEEP ALL CONDITIONS MATCHED BY SEEDS, BASE POLICY, AND ACTION BUDGET UNLESS EXPLICITLY STATED.

Condition	Episodes	Success	Recovery	Retries
Policy-only	TODO	TODO	–	–
Agentic retry	TODO	TODO	TODO	TODO
Agentic replan	TODO	TODO	TODO	TODO

B. Policy-Only Baseline

TODO summarize the current routed SmolVLA baseline and cite the local artifact paths in the camera-ready or supplement, not in the blinded submission.

C. Recovery Results

TODO compare `policy_only`, `agentic_retry`, and, if ready, `agentic_replan`. Report recovery-after-failure, retries per episode, latency, and failure categories.

V. Discussion

TODO explain when recovery helps, when it fails, and what this says about semantic reasoning around lightweight VLA policies.

VI. Limitations

TODO include limited task coverage, simulator-only evidence, verifier dependence, no weight updates, possible extra action budget, and no real robot claim unless real evidence is added.

VII. Conclusion

TODO one paragraph: what was tested, strongest measured result, and why the wrapper is useful as an evaluation and recovery layer for lightweight VLA policies.

References

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